

Figure 3

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Contents

1 Packages, data and functions	1
1.1 Packages	1
1.2 Data	2
2 Figure 3	2
2.1 Models	2
2.2 Figure	6
3 Versions of R and of the R-packages used	6

1 Packages, data and functions

1.1 Packages

```
library(tidyverse)
library(here)
library(survey)
library(lavaan)
library(lavaan.survey)
# lavaan.survey is an extension of lavaan that can deal with data independency.
# It is no longer on cran, but it is a trustworthy package (there is also a
# publication for it). The source code is
# here https://github.com/daob/lavaan.survey. To get the package you need to
# clone the repository linked above and then build the package locally.
```

1.2 Data

```
load(here("Data/3_ForecastsResults/3_MergedResults/complete.forecasts.RData"))

semDD.simplex <- complete.forecasts %>%
  dplyr::filter(forecast.method == "Simplex EDM",
               Target != "Total_biomass",
               Target != "Mean_oxygen") %>%
  ungroup()
```

2 Figure 3

2.1 Models

```
model <-
"
RMSE ~ median_richness + cv.res + pe5 + acf_lag5
cv.res ~ median_richness
pe5 ~ median_richness
acf_lag5 ~ median_richness
cv.res ~~ pe5
"

## create random effect variable
semDD.simplex$group <- paste0(semDD.simplex$Target, "_",
                             semDD.simplex$bottle, "_",
                             semDD.simplex$incubator)

# semDD.simplex, semDD.rf, semDD.mu, semDD.rnn, semDD.ar
out <- sem(model, data=semDD.simplex, group="light_treatment")

### Survey Design Object
des.card <- svydesign(ids = ~ group, nest=T, data = semDD.simplex)

### Post-Processing
```

```
fit.out.survey <- lavaan.survey(out, des.card, estimator = "MLM")
```

```
summary(fit.out.survey, standardized=T, rsquare=T)
```

```
## lavaan 0.6.17 ended normally after 105 iterations
```

```
##
```

```
## Estimator ML
```

```
## Optimization method NLMINB
```

```
## Number of model parameters 32
```

```
##
```

```
## Number of observations per group:
```

```
## Group 1 126
```

```
## Group 2 127
```

```
##
```

```
## Model Test User Model:
```

```
## Standard Scaled
```

```
## Test Statistic 3.219 3.270
```

```
## Degrees of freedom 4 4
```

```
## P-value (Chi-square) 0.522 0.514
```

```
## Scaling correction factor 0.984
```

```
## Satorra-Bentler correction
```

```
## Test statistic for each group:
```

```
## Group 1 2.977 3.024
```

```
## Group 2 0.242 0.246
```

```
##
```

```
## Parameter Estimates:
```

```
##
```

```
## Standard errors Robust.sem
```

```
## Information Expected
```

```
## Information saturated (h1) model Structured
```

```
##
```

```
##
```

```
## Group 1 [Group 1]:
```

```
##
```

```
## Regressions:
```

```
## Estimate Std.Err z-value P(>|z|) Std.lv Std.all
```

```

## RMSE ~
## median_richnss 0.015 0.031 0.469 0.639 0.015 0.035
## cv.res -0.602 0.177 -3.412 0.001 -0.602 -0.416
## pe5 0.085 0.659 0.128 0.898 0.085 0.017
## acf_lag5 -0.695 0.112 -6.230 0.000 -0.695 -0.368
## cv.res ~
## median_richnss 0.005 0.022 0.243 0.808 0.005 0.018
## pe5 ~
## median_richnss 0.010 0.007 1.459 0.145 0.010 0.117
## acf_lag5 ~
## median_richnss -0.018 0.019 -0.948 0.343 -0.018 -0.081
##
## Covariances:
## Estimate Std.Err z-value P(>|z|) Std.lv Std.all
## .cv.res ~~
## .pe5 -0.032 0.008 -4.110 0.000 -0.032 -0.784
##
## Intercepts:
## Estimate Std.Err z-value P(>|z|) Std.lv Std.all
## .RMSE 1.177 0.474 2.485 0.013 1.177 2.151
## .cv.res 0.416 0.119 3.486 0.000 0.416 1.101
## .pe5 0.595 0.036 16.347 0.000 0.595 5.471
## .acf_lag5 0.517 0.103 5.012 0.000 0.517 1.782
##
## Variances:
## Estimate Std.Err z-value P(>|z|) Std.lv Std.all
## .RMSE 0.203 0.041 5.006 0.000 0.203 0.678
## .cv.res 0.143 0.034 4.151 0.000 0.143 1.000
## .pe5 0.012 0.002 5.389 0.000 0.012 0.986
## .acf_lag5 0.084 0.006 13.688 0.000 0.084 0.993
##
## R-Square:
## Estimate
## RMSE 0.322
## cv.res 0.000
## pe5 0.014

```

```

##      acf_lag5          0.007
##
##
## Group 2 [Group 2]:
##
## Regressions:
##
##              Estimate  Std.Err  z-value  P(>|z|)  Std.lv  Std.all
##  RMSE ~
##      median_richnss   -0.077    0.035   -2.224    0.026   -0.077   -0.220
##      cv.res           -0.095    0.173   -0.548    0.583   -0.095   -0.072
##      pe5              0.430    0.569    0.756    0.449    0.430    0.092
##      acf_lag5         -0.350    0.143   -2.455    0.014   -0.350   -0.185
##  cv.res ~
##      median_richnss   -0.030    0.026   -1.150    0.250   -0.030   -0.111
##  pe5 ~
##      median_richnss    0.014    0.007    2.186    0.029    0.014    0.191
##  acf_lag5 ~
##      median_richnss    0.010    0.017    0.592    0.554    0.010    0.053
##
## Covariances:
##
##              Estimate  Std.Err  z-value  P(>|z|)  Std.lv  Std.all
##  .cv.res ~~
##      .pe5             -0.022    0.006   -3.968    0.000   -0.022   -0.628
##
## Intercepts:
##
##              Estimate  Std.Err  z-value  P(>|z|)  Std.lv  Std.all
##  .RMSE            1.055    0.394    2.681    0.007    1.055    2.235
##  .cv.res           0.713    0.143    4.978    0.000    0.713    1.982
##  .pe5              0.540    0.037   14.714    0.000    0.540    5.319
##  .acf_lag5         0.506    0.093    5.454    0.000    0.506    2.026
##
## Variances:
##
##              Estimate  Std.Err  z-value  P(>|z|)  Std.lv  Std.all
##  .RMSE            0.201    0.041    4.898    0.000    0.201    0.903
##  .cv.res           0.128    0.037    3.444    0.001    0.128    0.988
##  .pe5              0.010    0.001    8.040    0.000    0.010    0.963

```

```
##      .acf_lag5          0.062    0.007    9.524    0.000    0.062    0.997
##
## R-Square:
##              Estimate
##      RMSE          0.097
##      cv.res          0.012
##      pe5             0.037
##      acf_lag5        0.003
```

2.2 Figure

Figure 3 was done with MS PowerPoint and the above model.

3 Versions of R and of the R-packages used

R version: R version 4.3.2 (2023-10-31 ucrt).

Base packages: grid, stats, graphics, grDevices, utils, datasets, methods, base.

Other packages: lavaan.survey 1.1.3, lavaan 0.6-17, survey 4.2-1, survival 3.5-7, Matrix 1.6-5, here 1.0.1, lubridate 1.9.3, forcats 1.0.0, stringr 1.5.1, dplyr 1.1.4, purrr 1.0.2, readr 2.1.5, tidyr 1.3.1, tibble 3.2.1, ggplot2 3.4.4, tidyverse 2.0.0, knitr 1.45.

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